

AI ASSISTED CODING LAB - 7.3

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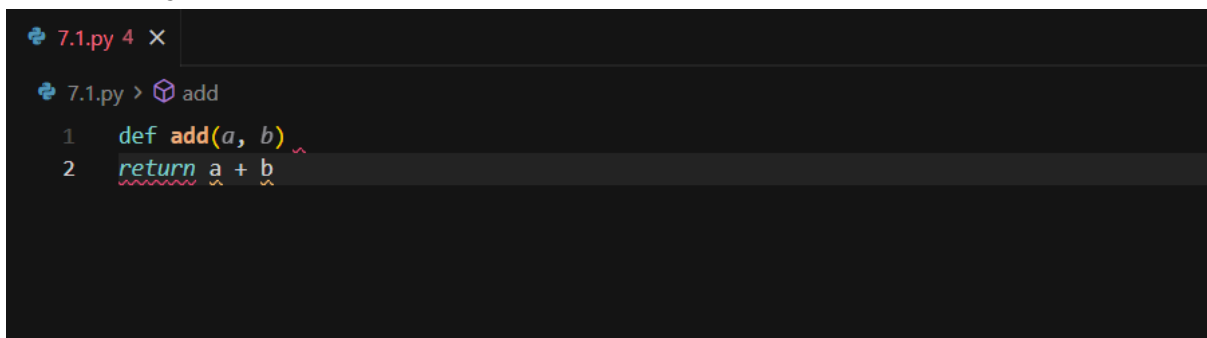
Batch : 41

Task 1: Fixing Syntax Errors

Scenario :

You are reviewing a Python program where a basic function definition contains a syntax error.

Code with syntax error



```
7.1.py 4 x
7.1.py > add
1 def add(a, b)
2 return a + b
```

Expected output:



```
Problems 4 Output Debug Console Terminal Ports
PS C:\Users\Hello\aiassistentcoding> & C:/Users/Hello/AppData/Local/Python/pythoncore
.py
File "c:\Users\Hello\aiassistentcoding\7.1.py", line 1
    def add(a, b)
                ^
SyntaxError: expected ':'
PS C:\Users\Hello\aiassistentcoding>
```

Error Identified by AI

Type of Error: SyntaxError

Reason: Python requires a colon (:) at the end of a function definition line.

Code debugged by ai

```
7.1.py
def add(a, b)
1  return a + b
2  def add(a, b):
3      return a + b
4
```

Output:

```
Problems Output Debug Console Terminal Ports
PS C:\Users\Hello\aiassistantcoding> & C:/Users/Hello/AppData/Local/Python/pythoncore-3.14-64/python
/Users/Hello/aiassistantcoding/7.1.py
PS C:\Users\Hello\aiassistantcoding> 2+4
6
PS C:\Users\Hello\aiassistantcoding>
```

Explanation by AI :

The AI detected a syntax error because the function declaration line did not end with a colon. In Python, the colon indicates the start of an indented code block. Without it, the interpreter cannot parse the function body.

- AI-generated explanation of the fix

Task 2: Debugging Logic Errors in Loops

Scenario

You are debugging a loop that runs infinitely due to a logical mistake.

Requirements

Code with errors

```
7.2.py > ...
1  i = 1
2  while i <= 10:
3      print(i)
4      i = i - 1
```

Error Identified by AI

Type of Error: Logical Error

Reason: The loop variable is decremented instead of incremented, so the condition $i \leq 10$ is always true.

Code debugged by ai

```
7.2.py > ...  
1 i = 1  
2 while i <= 10:  
3     print(i)  
4     i = i - 1  
5     i = i + 1  
#2303A52142
```

Output:

```
Problems Output Debug Console Terminal Ports  
PS C:\Users\Hello\aiassisstantcoding> & C:/Users/Hello/AppData/Local/Python/pythoncore-3.14-64/python.exe  
.py  
1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
PS C:\Users\Hello\aiassisstantcoding>
```

Explanation by AI

The loop condition depends on *i* increasing toward a stopping point. Decrementing *i* causes it to move away from the termination condition, resulting in an infinite loop.

Task 3: Handling Runtime Errors (Division by Zero)

Scenario

A function crashes due to division by zero.

Buggy code

```
7.3.py > ...  
1 def divide(a, b):  
2     return a / b  
3 print(divide(10, 0))
```

```
Problems Output Debug Console Terminal Ports
> & C:/Users/Hello/AppData/Local/Python/pythoncore-3.14-64/python.exe c:/Users/Hello/AppData/Local/Python/pythoncore-3.14-64/7.3.py
Traceback (most recent call last):
  File "c:\Users\Hello\aiassisstantcoding\7.3.py", line 3, in <module>
    print(divide(10, 0))
             ~~~~~~
  File "c:\Users\Hello\aiassisstantcoding\7.3.py", line 2, in divide
    return a / b
           ~~~~
ZeroDivisionError: division by zero
PS C:\Users\Hello\aiassisstantcoding>
```

Error Identified by AI

Type of Error: Runtime Error (ZeroDivisionError)

Reason: Division by zero is mathematically undefined.

Debugged code

```
7.3.py > ...
1 def divide(a, b):
2     try:
3         return a / b
4     except ZeroDivisionError:
5         print("Error: Division by zero is not allowed!")
6         return None
7 print(divide(10, 0))
8 print(divide(7, 0))
```

Output:

```
Problems Output Debug Console Terminal Ports
PS C:\Users\Hello\aiassisstantcoding> & C:/Users/Hello/AppData/Local/Python/pythoncore-3.14-64/python.exe c:/Users/Hello/AppData/Local/Python/pythoncore-3.14-64/7.3.py
Error: Division by zero is not allowed!
None
Error: Division by zero is not allowed!
None
PS C:\Users\Hello\aiassisstantcoding>
```

Explanation by AI

The AI added exception handling using try-except to prevent program termination. This ensures graceful handling of invalid input instead of crashing the program.

Task 4: Debugging Class Definition Errors

Scenario

A Python class constructor is incorrectly defined without the self parameter.

Buggy Code

```
7.4.py > ...  
1  def __init__(self, name, age):  
2  name = name  
3  age = age
```

Problems 3 Output Debug Console Terminal Ports

```
PS C:\Users\Hello\aiassisstantcoding> & C:/Users/Hello/AppData/Local/Python/pythoncore-3.14-64,  
.py  
File "c:\Users\Hello\aiassisstantcoding\7.4.py", line 2  
    name = name  
    ^^^^^  
IndentationError: expected an indented block after function definition on line 1  
PS C:\Users\Hello\aiassisstantcoding>
```

Error Identified by AI

Type of Error: TypeError / Object-Oriented Error

Reason: The self parameter is mandatory for instance variable access.

Debugged code

```
7.4.py > _init_  
1  def __init__(self, name, age):  
2      self.name = name  
3      self.age = age
```

Explanation by AI

The self parameter represents the current object instance. Without it, instance variables cannot be properly assigned or accessed.

Task 5: Resolving Index Errors in Lists

Scenario

A program crashes when accessing an invalid list index.

Buggy Code

```
7.5.py > ...  
1  numbers = [10, 20, 30, 40, 50]  
2  print(numbers[8])
```

```
Problems Output Debug Console Terminal Ports
PS C:\Users\Hello\aiassistantcoding> & C:/Users/Hello/AppData/Local/Python/pythoncore-3.14-64/python
assistantcoding/7.5.py
Traceback (most recent call last):
  File "c:\Users\Hello\aiassistantcoding\7.5.py", line 2, in <module>
    print(numbers[8])
                ~~~~^~
IndexError: list index out of range
PS C:\Users\Hello\aiassistantcoding>
```

Error Identified by AI

Type of Error: IndexError

Reason: Index is outside the valid range of the list.

Debugged code

```
7.5.py > ...
1 numbers = [10, 20, 30, 40, 50]
2 try:
3     print(numbers[8])
4 except IndexError:
5     print("Index out of range")
6 except ValueError:
7     print("Invalid input")
8 except Exception as e:
9     print(e)
10 finally:
11     print("This will always execute")
```

output:

```
Problems Output Debug Console Terminal Ports Python
PS C:\Users\Hello\aiassistantcoding> & C:/Users/Hello/AppData/Local/Python/pythoncore-3.14-64/pyt
/aiassistantcoding/7.5.py
Index out of range
This will always execute
PS C:\Users\Hello\aiassistantcoding>
```

Explanation by AI

The AI suggested two safe approaches: checking the index against list length or handling the exception using try-except. Both prevent program crashes.